

Principles of Programming Languages

Lecture 9: Structural Operational Semantics. Big-step and Small-step SOS.

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Thank you...

... Grigore Roşu for the content of this presentation and the free access to the book chapters.

Outline

Motivation

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IMP

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Big-step SOS

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Big-step SOS

Small-step SOS

Motivation

- ▶ A language designer should understand the existing design approaches, techniques and tools.

IMP-syntax

```
AExp ::= Int | Id  
      | AExp "/" AExp  
      > AExp "+" AExp  
      | "(" AExp ")"
```

```
BExp ::= Bool  
      | "!" BExp  
      | BExp "and" BExp  
      | AExp "<=" AExp  
      | "(" BExp ")"
```

```
Block ::= "{" "}"  
       | "{" Stmt "}"
```

```
Stmt ::= Block  
      | Id "=" AExp ";"  
      | "if" "(" BExp ")" Stmt "else" Stmt  
      | "while" "(" BExp ")" Stmt  
      > Stmt Stmt
```

```
Pgm ::= "int" Ids ";" Stmt  
Ids ::= List{Id, ",", "}"
```

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$$\sigma : Id \rightarrow Int$$

- ▶ Example: $\sigma = x \mapsto 10, y \mapsto -1$
- ▶ We denote by *State* the set of such functions:

$$[Id \rightarrow Int]^{finite}$$

State operations

- ▶ *Lookup*

$$_(_) : \textit{State} \times \textit{Id} \rightarrow \textit{Int}$$

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- ▶ *Initialization*

$$_ \mapsto _ : \textit{List}\{\textit{Id}\} \times \textit{Int} \rightarrow \textit{State}$$

Big-step SOS

Ingredients:

- ▶ *Configuration*
 - ▶ Tuple containing semantic ingredients
 - ▶ Examples: $\langle a_1, \sigma \rangle$, $\langle a_1 + a_2, \sigma \rangle$, $\langle i_1 \rangle$, $\langle i_1 +_{Int} i_2 \rangle$, ...

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 - ▶ Example: **premises**, **conclusion**, and **evaluates to**

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Rules

- General form: **premises**, **conclusion**, and **evaluates to**

$$\frac{C_1 \Downarrow R_1 \quad C_2 \Downarrow R_2 \quad \cdots \quad C_n \Downarrow R_n}{C_0 \Downarrow R_0}$$

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BIGSTEP-SEQ:
$$\frac{\langle s_1, \sigma \rangle \Downarrow \langle \sigma_1 \rangle \quad \langle s_2, \sigma_1 \rangle \Downarrow \langle \sigma_2 \rangle}{\langle s_1 \ s_2, \sigma \rangle \Downarrow \langle \sigma_2 \rangle}$$

Big-step rules for IMP – 4: statements continued

BIGSTEP-IFTRUE:

$$\frac{\langle b, \sigma \rangle \Downarrow \langle \text{true} \rangle \quad \langle s_1, \sigma \rangle \Downarrow \langle \sigma_1 \rangle}{\langle \text{if } (b) \ s_1 \text{ else } s_2, \sigma \rangle \Downarrow \langle \sigma_1 \rangle}$$

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BIGSTEP-WHILEFALSE:

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$$\text{BIGSTEP-INIT:} \quad \frac{\langle s, x! \mapsto 0 \rangle \Downarrow \langle \sigma \rangle}{\langle \text{int } x!; s \rangle \Downarrow \langle \sigma \rangle}$$

Big-step rules examples

- Recall BIGSTEP-DIV:

$$\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1/a_2, \sigma \rangle \Downarrow \langle i_1/\text{Int } i_2 \rangle} \text{ if } i_2 \neq 0$$

- The following is a correct instance of BIGSTEP-DIV:

$$\frac{\langle x, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 10 \rangle \quad \langle 2, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 2 \rangle}{\langle x/2, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 5 \rangle}$$

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- What about the following?

$$\frac{\langle x, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 8 \rangle \quad \langle 2, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 2 \rangle}{\langle x/2, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 4 \rangle}$$

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- Is the following instance correct w.r.t. BIGSTEP-DIV?

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- Answer: **no**

Derivation example

We assume that $\sigma \stackrel{\text{def}}{=} (x \mapsto 8, y \mapsto 0)$:

$$\frac{\text{BIGSTEP-LOOKUP} \frac{\cdot}{\langle x, \sigma \rangle \Downarrow \langle 8 \rangle} \quad \text{BIGSTEP-DIV} \frac{\text{BIGSTEP-LOOKUP} \frac{\cdot}{\langle y, \sigma \rangle \Downarrow \langle 0 \rangle} \quad \text{BIGSTEP-LOOKUP} \frac{\cdot}{\langle x, \sigma \rangle \Downarrow \langle 8 \rangle} \quad \text{BIGSTEP-LOOKUP} \frac{\cdot}{\langle 2, \sigma \rangle \Downarrow \langle 2 \rangle}}{\langle y/x, \sigma \rangle \Downarrow \langle 0 \rangle} \quad \text{BIGSTEP-INT} \quad \text{BIGSTEP-ADD} \quad \text{BIGSTEP-DIV} \frac{\langle x/(y/x + 2), \sigma \rangle \Downarrow \langle 4 \rangle}}{\langle y/x + 2, \sigma \rangle \Downarrow \langle 2 \rangle}}$$

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Rules

- ▶ General form:

$$\frac{C_1 \rightarrow C'_1 \quad C_2 \rightarrow C'_2 \quad \cdots \quad C_n \rightarrow C'_n}{C_0 \rightarrow C'_0}$$

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Note: + is non-deterministic!

Small-step rules for IMP – 2: arithmetic expressions

$$\text{SMALLSTEP-DIV1: } \frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1/a_2, \sigma \rangle \rightarrow \langle a'_1/a_2, \sigma \rangle}$$

Small-step rules for IMP – 2: arithmetic expressions

$$\text{SMALLSTEP-DIV1: } \frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1/a_2, \sigma \rangle \rightarrow \langle a'_1/a_2, \sigma \rangle}$$

$$\text{SMALLSTEP-DIV2: } \frac{\langle a_2, \sigma \rangle \rightarrow \langle a'_2, \sigma \rangle}{\langle a_1/a_2, \sigma \rangle \rightarrow \langle a_1/a'_2, \sigma \rangle}$$

Small-step rules for IMP – 2: arithmetic expressions

$$\text{SMALLSTEP-DIV1: } \frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1/a_2, \sigma \rangle \rightarrow \langle a'_1/a_2, \sigma \rangle}$$

$$\text{SMALLSTEP-DIV2: } \frac{\langle a_2, \sigma \rangle \rightarrow \langle a'_2, \sigma \rangle}{\langle a_1/a_2, \sigma \rangle \rightarrow \langle a_1/a'_2, \sigma \rangle}$$

$$\text{SMALLSTEP-DIV: } \langle i_1/i_2, \sigma \rangle \rightarrow \langle i_1/\text{Int } i_2, \sigma \rangle \quad \text{if } i_2 \neq 0$$

Small-step rules for IMP – 2: arithmetic expressions

$$\text{SMALLSTEP-DIV1: } \frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1/a_2, \sigma \rangle \rightarrow \langle a'_1/a_2, \sigma \rangle}$$

$$\text{SMALLSTEP-DIV2: } \frac{\langle a_2, \sigma \rangle \rightarrow \langle a'_2, \sigma \rangle}{\langle a_1/a_2, \sigma \rangle \rightarrow \langle a_1/a'_2, \sigma \rangle}$$

$$\text{SMALLSTEP-DIV: } \langle i_1/i_2, \sigma \rangle \rightarrow \langle i_1/\text{Int } i_2, \sigma \rangle \quad \text{if } i_2 \neq 0$$

Note: / is also non-deterministic!

Small-step rules for IMP – 3: boolean expressions

$$\text{SMALLSTEP-LEQ1: } \frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1 \leq a_2, \sigma \rangle \rightarrow \langle a'_1 \leq a_2, \sigma \rangle}$$

Small-step rules for IMP – 3: boolean expressions

$$\text{SMALLSTEP-LEQ1: } \frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1 \leq a_2, \sigma \rangle \rightarrow \langle a'_1 \leq a_2, \sigma \rangle}$$

$$\text{SMALLSTEP-LEQ2: } \frac{\langle a_2, \sigma \rangle \rightarrow \langle a'_2, \sigma \rangle}{\langle i_1 \leq a_2, \sigma \rangle \rightarrow \langle i_1 \leq a'_2, \sigma \rangle}$$

Small-step rules for IMP – 3: boolean expressions

$$\text{SMALLSTEP-LEQ1: } \frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1 \leq a_2, \sigma \rangle \rightarrow \langle a'_1 \leq a_2, \sigma \rangle}$$

$$\text{SMALLSTEP-LEQ2: } \frac{\langle a_2, \sigma \rangle \rightarrow \langle a'_2, \sigma \rangle}{\langle i_1 \leq a_2, \sigma \rangle \rightarrow \langle i_1 \leq a'_2, \sigma \rangle}$$

$$\text{SMALLSTEP-LEQ: } \langle i_1 \leq i_2, \sigma \rangle \rightarrow \langle i_1 \leq_{Int} i_2, \sigma \rangle$$

Small-step rules for IMP – 3: boolean expressions

$$\text{SMALLSTEP-LEQ1: } \frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1 \leq a_2, \sigma \rangle \rightarrow \langle a'_1 \leq a_2, \sigma \rangle}$$

$$\text{SMALLSTEP-LEQ2: } \frac{\langle a_2, \sigma \rangle \rightarrow \langle a'_2, \sigma \rangle}{\langle i_1 \leq a_2, \sigma \rangle \rightarrow \langle i_1 \leq a'_2, \sigma \rangle}$$

$$\text{SMALLSTEP-LEQ: } \langle i_1 \leq i_2, \sigma \rangle \rightarrow \langle i_1 \leq_{Int} i_2, \sigma \rangle$$

Note: i_1 is used to ensure sequential strictness.

Small-step rules for IMP – 4: boolean expressions

SMALLSTEP-NOT:

$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle !b, \sigma \rangle \rightarrow \langle !b', \sigma \rangle}$$

Small-step rules for IMP – 4: boolean expressions

SMALLSTEP-NOT:
$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle !b, \sigma \rangle \rightarrow \langle !b', \sigma \rangle}$$

SMALLSTEP-NOTTRUE:
$$\langle !\text{true}, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

Small-step rules for IMP – 4: boolean expressions

SMALLSTEP-NOT:
$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle !b, \sigma \rangle \rightarrow \langle !b', \sigma \rangle}$$

SMALLSTEP-NOTTRUE:
$$\langle !\text{true}, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

SMALLSTEP-NOTTRUE:
$$\langle !\text{false}, \sigma \rangle \rightarrow \langle \text{true}, \sigma \rangle$$

Small-step rules for IMP – 4: boolean expressions

SMALLSTEP-NOT:
$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle !b, \sigma \rangle \rightarrow \langle !b', \sigma \rangle}$$

SMALLSTEP-NOTTRUE:
$$\langle !\text{true}, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

SMALLSTEP-NOTTRUE:
$$\langle !\text{false}, \sigma \rangle \rightarrow \langle \text{true}, \sigma \rangle$$

SMALLSTEP-AND1:
$$\frac{\langle b_1, \sigma \rangle \rightarrow \langle b'_1, \sigma \rangle}{\langle b_1 \text{ and } b_2, \sigma \rangle \rightarrow \langle b'_1 \text{ and } b_2, \sigma \rangle}$$

Small-step rules for IMP – 4: boolean expressions

$$\text{SMALLSTEP-NOT:} \quad \frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle !b, \sigma \rangle \rightarrow \langle !b', \sigma \rangle}$$

$$\text{SMALLSTEP-NOTTRUE:} \quad \langle !\text{true}, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

$$\text{SMALLSTEP-NOTTRUE:} \quad \langle !\text{false}, \sigma \rangle \rightarrow \langle \text{true}, \sigma \rangle$$

$$\text{SMALLSTEP-AND1:} \quad \frac{\langle b_1, \sigma \rangle \rightarrow \langle b'_1, \sigma \rangle}{\langle b_1 \text{ and } b_2, \sigma \rangle \rightarrow \langle b'_1 \text{ and } b_2, \sigma \rangle}$$

$$\text{SMALLSTEP-ANDFALSE:} \quad \langle \text{false and } b_2, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

Small-step rules for IMP – 4: boolean expressions

$$\text{SMALLSTEP-NOT:} \quad \frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle !b, \sigma \rangle \rightarrow \langle !b', \sigma \rangle}$$

$$\text{SMALLSTEP-NOTTRUE:} \quad \langle !\text{true}, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

$$\text{SMALLSTEP-NOTTRUE:} \quad \langle !\text{false}, \sigma \rangle \rightarrow \langle \text{true}, \sigma \rangle$$

$$\text{SMALLSTEP-AND1:} \quad \frac{\langle b_1, \sigma \rangle \rightarrow \langle b'_1, \sigma \rangle}{\langle b_1 \text{ and } b_2, \sigma \rangle \rightarrow \langle b'_1 \text{ and } b_2, \sigma \rangle}$$

$$\text{SMALLSTEP-ANDFALSE:} \quad \langle \text{false and } b_2, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

$$\text{SMALLSTEP-ANDTRUE:} \quad \langle \text{true and } b_2, \sigma \rangle \rightarrow \langle b_2, \sigma \rangle$$

Example

Assume that $\sigma(x) = 0$:

$$\begin{array}{c} \cdot \\ \hline \langle x, \sigma \rangle \rightarrow \langle 1, \sigma \rangle \quad \text{SMALLSTEP-LOOKUP} \\ \hline \langle (y/x), \sigma \rangle \rightarrow \langle (y/1), \sigma \rangle \quad \text{SMALLSTEP-DIV2} \\ \hline \langle (x+(y/x)), \sigma \rangle \rightarrow \langle (x+(y/1)), \sigma \rangle \quad \text{SMALLSTEP-ADD2} \\ \hline \langle (x+(y/x)) \leq x, \sigma \rangle \rightarrow \langle (x+(y/1)) \leq x, \sigma \rangle \quad \text{SMALLSTEP-LEQ1} \end{array}$$

Small-step rules for IMP – 5: statements

SMALLSTEP-BLOCK: $\langle \{\mathbf{s}\}, \sigma \rangle \rightarrow \langle \mathbf{s}, \sigma \rangle$

Small-step rules for IMP – 5: statements

SMALLSTEP-BLOCK: $\langle \{s\}, \sigma \rangle \rightarrow \langle s, \sigma \rangle$

SMALLSTEP-ASSIGN2:
$$\frac{\langle a, \sigma \rangle \rightarrow \langle a', \sigma \rangle}{\langle x = a; , \sigma \rangle \rightarrow \langle x = a'; , \sigma \rangle}$$

Small-step rules for IMP – 5: statements

SMALLSTEP-BLOCK: $\langle \{s\}, \sigma \rangle \rightarrow \langle s, \sigma \rangle$

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$$\frac{\langle a, \sigma \rangle \rightarrow \langle a', \sigma \rangle}{\langle x = a; , \sigma \rangle \rightarrow \langle x = a'; , \sigma \rangle}$$

SMALLSTEP-ASSIGN: $\langle x = i; , \sigma \rangle \rightarrow \langle \{ \}, \sigma[i/x] \rangle \quad \text{if } \sigma(x) \neq \perp$

Small-step rules for IMP – 5: statements

SMALLSTEP-BLOCK: $\langle \{s\}, \sigma \rangle \rightarrow \langle s, \sigma \rangle$

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SMALLSTEP-ASSIGN: $\langle x = i; , \sigma \rangle \rightarrow \langle \{ \}, \sigma[i/x] \rangle \quad \text{if } \sigma(x) \neq \perp$

SMALLSTEP-SEQ1:
$$\frac{\langle s_1, \sigma \rangle \rightarrow \langle s'_1, \sigma' \rangle}{\langle s_1 \ s_2, \sigma \rangle \rightarrow \langle s'_1 \ s_2, \sigma' \rangle}$$

Small-step rules for IMP – 5: statements

SMALLSTEP-BLOCK: $\langle \{s\}, \sigma \rangle \rightarrow \langle s, \sigma \rangle$

SMALLSTEP-ASSIGN2:
$$\frac{\langle a, \sigma \rangle \rightarrow \langle a', \sigma \rangle}{\langle x = a; , \sigma \rangle \rightarrow \langle x = a'; , \sigma \rangle}$$

SMALLSTEP-ASSIGN: $\langle x = i; , \sigma \rangle \rightarrow \langle \{ \}, \sigma[i/x] \rangle \quad \text{if } \sigma(x) \neq \perp$

SMALLSTEP-SEQ1:
$$\frac{\langle s_1, \sigma \rangle \rightarrow \langle s'_1, \sigma' \rangle}{\langle s_1 \ s_2, \sigma \rangle \rightarrow \langle s'_1 \ s_2, \sigma' \rangle}$$

SMALLSTEP-BLOCK: $\langle \{ \} \ s_2, \sigma \rangle \rightarrow \langle s_2, \sigma \rangle$

Small-step rules for IMP – 6: statements

$$\text{SMALLSTEP-IF1:} \quad \frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \text{if } (b) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle \text{if } (b') \ s_1 \text{ else } s_2, \sigma \rangle}$$

Small-step rules for IMP – 6: statements

$$\text{SMALLSTEP-IF1:} \quad \frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \text{if } (b) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle \text{if } (b') \ s_1 \text{ else } s_2, \sigma \rangle}$$

$$\text{SMALLSTEP-IFTRUE:} \quad \langle \text{if } (\text{true}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_1, \sigma \rangle$$

Small-step rules for IMP – 6: statements

$$\text{SMALLSTEP-IF1:} \quad \frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \text{if } (b) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle \text{if } (b') \ s_1 \text{ else } s_2, \sigma \rangle}$$

$$\text{SMALLSTEP-IFTRUE:} \quad \langle \text{if } (\text{true}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_1, \sigma \rangle$$

$$\text{SMALLSTEP-IFFALSE:} \quad \langle \text{if } (\text{false}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_2, \sigma \rangle$$

Small-step rules for IMP – 6: statements

SMALLSTEP-IF1:
$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \text{if } (b) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle \text{if } (b') \ s_1 \text{ else } s_2, \sigma \rangle}$$

SMALLSTEP-IFTRUE:
$$\langle \text{if } (\text{true}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_1, \sigma \rangle$$

SMALLSTEP-IFFALSE:
$$\langle \text{if } (\text{false}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_2, \sigma \rangle$$

SMALLSTEP-WHILE:
$$\langle \text{while } (b) \ s, \sigma \rangle \rightarrow \langle \text{if } (b) \ \{s \text{ while } (b) \ s\} \text{ else } \{\}, \sigma \rangle$$

Small-step rules for IMP – 6: statements

SMALLSTEP-IF1:
$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \text{if } (b) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle \text{if } (b') \ s_1 \text{ else } s_2, \sigma \rangle}$$

SMALLSTEP-IFTRUE:
$$\langle \text{if } (\text{true}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_1, \sigma \rangle$$

SMALLSTEP-IFFALSE:
$$\langle \text{if } (\text{false}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_2, \sigma \rangle$$

SMALLSTEP-WHILE:
$$\langle \text{while } (b) \ s, \sigma \rangle \rightarrow \langle \text{if } (b) \{s \text{ while } (b) \ s\} \text{ else } \{\}, \sigma \rangle$$

SMALLSTEP-INIT:
$$\langle \text{int } x!; \ s, \sigma \rangle \rightarrow \langle s, x! \mapsto 0 \rangle$$

Resources

1. Main course page:

[http://fsl.cs.illinois.edu/index.php/CS422_-_Programming_Language_Design_\(Spring_2015\)](http://fsl.cs.illinois.edu/index.php/CS422_-_Programming_Language_Design_(Spring_2015))

2. BigStep-SOS:

<http://fsl.cs.illinois.edu/images/c/cf/CS422-Spring-2015-02b-BigStep.pdf>

3. SmallStep-SOS:

<http://fsl.cs.illinois.edu/images/6/65/CS422-Spring-2015-02c-SmallStep.pdf>